

# MOHAB MOHAMED RABIE MOHAMED

## AI & Machine Learning Engineer

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## PROFESSIONAL SUMMARY

AI and Machine Learning Engineer holding a B.Sc. in Artificial Intelligence (Excellent with Honors) from Helwan University and currently pursuing postgraduate studies in AI at Cairo University. Hands-on experience architecting and deploying end-to-end Machine Learning, Computer Vision, NLP, and LLM solutions using PyTorch, TensorFlow, FastAPI, and FAISS. Over 3 years of experience instructing Python and Machine Learning, demonstrating strong software fundamentals and technical communication.

## EDUCATION

**Cairo University — Faculty of Computers and Artificial Intelligence** Cairo, Egypt  
Present  
*Postgraduate Studies in Artificial Intelligence*

**Helwan University — Faculty of Computers and Artificial Intelligence** Cairo, Egypt  
2025  
*B.Sc. in Computers and Artificial Intelligence (Major: AI)*

- **Grade:** Excellent with Honors
- **Graduation Project:** FITAI – AI-Powered Fitness & Nutrition Platform (Grade: A+)

## AI & MACHINE LEARNING PROJECTS

**Youth Entities AI Chatbot — Engineers for Sustainable Egypt**  
*Python, FastAPI, LLaMA, RAG, FAISS, JavaScript*

- Engineered an end-to-end Retrieval-Augmented Generation (RAG) pipeline utilizing FAISS for vector search and LLaMA for precise, contextual query response generation.
- Built scalable RESTful API endpoints using FastAPI to deliver response payloads to the web frontend with low latency.

**FITAI — AI-Powered Fitness & Nutrition Platform | Graduation Project (Grade: A+)**  
*Python, PyTorch, YOLOv8, MediaPipe, OpenCV, FastAPI, Flask, React Native*

- Architected a multi-modal AI platform integrating pose estimation, computer vision, and conversational AI into cross-platform web and mobile applications.
- Implemented real-time exercise form analysis and repetition tracking using MediaPipe joint tracking and pose estimation algorithms.
- Fine-tuned a YOLOv8 object detection model for real-time food item recognition and automated calorie estimation.
- Deployed core model pipelines behind REST APIs using FastAPI and Flask to ensure modular integration across React and React Native frontends.

## TECHNICAL EXPERIENCE

**Artificial Intelligence & Python Instructor** Cairo, Egypt  
*IBI Institute*

- Taught Machine Learning, AI algorithms, and Python programming to 100+ students through project-based coursework.

- Designed curriculum modules covering regression, classification, computer vision basics, and neural network architectures.

### Programming Instructor

Cairo, Egypt

3C

- Delivered hands-on training in algorithmic thinking, data structures, and core programming paradigms.

### Online Artificial Intelligence Instructor

Remote

Steam Park UAE

- Conducted live online training sessions covering applied machine learning and practical Python implementations.

## TECHNICAL SKILLS

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**Languages:** Python, C++, SQL, JavaScript

**Machine Learning:** Scikit-learn, Regression, Classification, Clustering, Feature Engineering, Model Evaluation

**Deep Learning & Frameworks:** PyTorch, TensorFlow, CNNs, Transfer Learning

**Computer Vision:** OpenCV, YOLOv8, MediaPipe, Object Detection

**NLP & GenAI:** Transformers, LLMs, RAG, Embeddings, FAISS, LLaMA

**Backend & Web:** FastAPI, Flask, REST APIs, React, React Native, HTML5, CSS3

**Developer Tools:** Git, GitHub, Jupyter Notebook, Google Colab, VS Code

## LANGUAGES

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**Arabic:** Native | **English:** Advanced (C1) | **French:** Elementary (A2)